

Lab 5: Variables
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SYNTACTICAL GRAMMAR DESIGN

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```
program      -> statement* EOF ;

statement    -> printStmt
              | varDecl
              | block
              | expressionStmt ;

printStmt    -> "print" expression ";" ;

varDecl      -> "new" IDENTIFIER ("=" expression)? ";" ;

block        -> "{" statement* "}" ;

expressionStmt -> expression ";" ;

expression   -> assignment ;

assignment   -> IDENTIFIER "=" assignment
              | equality ;

equality     -> comparison ( ( "==" | "!=" ) comparison ) * ;

comparison   -> term ( ( ">" | ">=" | "<" | "<=" ) term ) * ;

term         -> factor ( ( "+" | "-" ) factor ) * ;

factor       -> unary ( ( "*" | "/" ) unary ) * ;

unary        -> ( "-" | "!" ) unary
              | primary ;

primary      -> NUMBER
              | STRING
              | "TRUE"
              | "FALSE"
              | "NULL"
              | IDENTIFIER
              | "(" expression ")" ;
```

Variable declaration, assignment, and access, with print statement:

```
===== SLOTH SHELL =====  
> new x = 4;  
> print x;  
4  
> x = NULL;  
> print x;  
NULL  
> x = TRUE;  
> print x;  
TRUE  
> x = FALSE;  
> print x;  
FALSE  
> 
```

Input:

Output:

```
new a = "global a";  
new b = "global b";  
new c = "global c";  
{  
    new a = "outer a";  
    new b = "outer b";  
    {  
        new a = "inner a";  
        print a;  
        print b;  
        print c;  
    }  
    print a;  
    print b;  
    print c;  
}  
print a;  
print b;  
print c;
```

```
inner a  
outer b  
global c  
outer a  
outer b  
global c  
global a  
global b  
global c
```

Missing semicolon	<pre>> new a = 5 [line 1] Error at end: Expect ';' after variable declaration. [Parse Error at '' (line 1)] Expect ';' after variable declaration.</pre>
Invalid assignment target	<pre>> 5 = a; [line 1] Error at '=': Invalid assignment target. [Parse Error at '=' (line 1)] Invalid assignment target.</pre>
Undefined variable access	<pre>> print b; [Runtime Error] Undefined variable 'b'.</pre>
Unmatched braces in block	<pre>> { [line 1] Error at end: Expect '}' after block. [Parse Error at '' (line 1)] Expect '}' after block.</pre>
Invalid expressions	<pre>> new x =; [line 1] Error at ';': Expect expression. [Parse Error at ';' (line 1)] Expect expression.</pre>
Runtime errors: type mismatch	<pre>> new str = "hello"; > str = str - 1; [Runtime Error] Operands must be numbers for '-'</pre>
Division by zero	<pre>> print 10/0; [Runtime Error] Division by zero</pre>